

Recursion is a case where we solve the problem by dividing the problem into sub-problems and keep repeating that until we hit a base-case and then returning and accumulating the solution as it returns back to the original caller. [Please practice recursion from CSE220 if you forgot. Solving some basic recursion problem will suffice also.

The handwritten note for this time complexity analysis for different recursive scenarios are given in this pdf:

[Recursion time complexity analysis](#)

[OPTIONAL FOR FURTHER STUDY IF YOU REALLY WANT]

The 3rd approach named "Master Theorem" is formulated and the mathematical derivation of that theorem can be found here:

[Recursive time complexity note Stanford](#)

You can see these lecture classes from MIT if you really want to:

[MIT Algo Lecture 2 Fall 2005](#)

[MIT Algo Lecture 3 Fall 2005](#)

Also, These may be very complicated and redundant for some people. You can also watch this playlist [Video 2 to 2.5]

[Abdullah Bari lecture series](#)